

November 25, 2025

1. We study an electron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 25 ns; after this time the particle decays. The average lifetime measured in the reference frame attached to the particles is 7.5 ns . What is the propagation velocity of the particle beam?
2. A certain bacterium doubles its population in 25 days. Two such bacteria are placed in a spaceship and sent on a 800-day space journey. During the trip, the spaceship moves at a constant speed of 0.995 c . How many bacteria will be on board at the moment of return?
3. A particle moves with a velocity such that $\frac{v}{c} = 0.9999$. Determine the value of $\frac{m}{m_0}$!
4. Determine the relativistic mass and the velocity of an electron whose kinetic energy is 1 MeV ($1.6 \cdot 10^{-13}\text{ J}$). The rest mass of the electron is $9.11 \cdot 10^{-31}\text{ kg}$.
5. Calculate the rest energy of a proton, corresponding to its rest mass of $1.67 \cdot 10^{-27}\text{ kg}$, in joules and in electronvolts! ($1\text{ MeV} = 1.6 \cdot 10^{-13}\text{ J}$)

Dr. Gabor FACSKO
facsko.gabor@uni-obuda.hu